The two T^2 gerbe ends are coupled, not independent: a strict one-way forced correlation $\delta \Rightarrow [\mathfrak{g}]$

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Abstract

For a structure group K of nilpotency class ≤ 2, the nonabelian inventory gerbe on the torus $T^2 = \mathrm{B}\mathbb{Z}^2$ carries two obstruction classes sharing the hinge $\mathrm{Inn}K$: the *strictification* class $\delta(\rho) = [A, B] \in H^2(T^2; \mathrm{Inn}K)$ and the *central* class $[\mathfrak{g}] \in H^2(T^2; Z(K))$. We prove they are *not* independent. They satisfy a forced one-way correlation

$$\delta \text{ couples on } T^2 \implies [\mathfrak{g}] \text{ couples on } T^2,$$

so the corner ($\delta \neq 0$, $[\mathfrak{g}] = 0$) is *forced empty*. The cause is a structural asymmetry between two abelian-kernel extensions: the central extension $1 \to Z(K) \to K \to \mathrm{Inn}K \to 1$ is non-split for every nonabelian class-2 K and carries the *trivial* $\mathrm{Inn}K$-action, so $[\mathfrak{g}]$ universally couples, while the strictification extension $\mathrm{Aut}K \to \mathrm{Out}K$ *can* split, so δ is selective. The implication is *strict*: the converse fails at $K = D_4$ ($[\mathfrak{g}] \neq 0$ while $\delta \equiv 0$), the dihedral $4 \mid n$ witness at $n = 4$. This is the T^2-analogue of the S^1-floor $D1 \Rightarrow D2$ dichotomy. A battery cross-tabulation confirms the empty corner.

1 Verdict

Over the base $T^2 = \mathrm{B}\mathbb{Z}^2$ we decide, for the two Giraud obstruction stages of the inventory gerbe, between

- **(A)** *independence* — both off-diagonal corners ($\delta \neq 0, [\mathfrak{g}] = 0$) and ($\delta = 0, [\mathfrak{g}] \neq 0$) realized; and
- **(B)** *forced correlation* — some implication always holds.

The answer is **(B)**. For every finite group K of nilpotency class ≤ 2 (the domain on which both obstructions take the clean abelian-H^2 form; the implication extends to all finite nilpotent K, Remark 5),

$$\boxed{\delta \text{ couples on } T^2 \implies [\mathfrak{g}] \text{ couples on } T^2}$$

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i.e. the corner ($\delta \neq 0,\ [\mathfrak{g}] = 0$) is *forced empty*. The converse corner ($\delta = 0,\ [\mathfrak{g}] \neq 0$) *is* occupied (e.g. $K = D_4$, the dihedral $4 \mid n$ witness at $n = 4$), so the correlation is *strict and one-directional*. There is no logical independence.

The group-theoretic cause is a structural asymmetry between the two extensions that the symmetric picture — "both are commutator obstructions of an abelian-kernel extension through the hinge Inn$K$" — hides:

- the *central* extension $1 \to Z(K) \to K \to \mathrm{Inn}K \to 1$ is *non-split for every nonabelian* class-2 K, and its coefficient $Z(K)$ carries the *trivial* Inn$K$-action, so $[\mathfrak{g}]$ is *universally obstructed* — every nonabelian class-2 group couples at the central end;
- the *strictification* extension $1 \to \mathrm{Inn}K \to \mathrm{Aut}K \to \mathrm{Out}K \to 1$ *can split* (it does for $D_4, Q_8, \mathrm{Heis}(\mathbb{F}_3), M_{16}, C_4 \rtimes C_4$), killing δ *selectively*.

Strictification coupling is therefore a strictly stronger non-commutativity condition than central coupling, and it cannot occur without the weaker one.

What is really being said (a temperance note). For nilpotent K the central end is elementary: $[\mathfrak{g}]$ *couples* $\iff$ K *nonabelian* (Lemma 2 and Remark 5). So the headline implication reduces to "δ couples $\Rightarrow$ K nonabelian" (Lemma 3, elementary). In other words the central end merely *detects nonabelianness*; it carries no information beyond that. The genuine mathematical content — and the hard, unresolved part — is not the correlation itself but *which* nonabelian groups δ-couple at all. That is an Aut-structure question (does (Σ) admit a homomorphic section?), and the cross-tab shows it is delicate: most nonabelian groups do *not* δ-couple. We therefore present (B) not as a deep coupling law but as the observation that one end is a coarse nonabelianness detector dominated by the other.

2 Setup: the two ends as one lemma, applied twice

Let K be finite of nilpotency class ≤ 2, so $[K,K] \subseteq Z(K)$ and $M := \mathrm{Inn}K = K/Z(K)$ is *abelian*. Two abelian-kernel group extensions share the hinge Inn$K$:

$$(\Sigma)\quad 1 \to \mathrm{Inn}K \to \mathrm{Aut}K \xrightarrow{\pi} \mathrm{Out}K \to 1, \qquad (Z)\quad 1 \to Z(K) \to K \xrightarrow{\mathrm{ad}} \mathrm{Inn}K \to 1.$$

Over the base $T^2 = \mathrm{B}\mathbb{Z}^2$ (nerve of $\mathcal{C} = \mathrm{B}\mathbb{Z}^2$), the proved δ-commutator lemma (`coupled-T2-general-delta-lemma`) computes the degree-2 connecting/lifting obstruction of *any* such extension $1 \to A \to E \to Q \to 1$ (A abelian) for a band $b : \mathbb{Z}^2 \to Q$: with lifts $x, y \in E$ of $b(s), b(t)$,

$$\mathrm{obs}(b) = \big[\,[x,y]\,\big] \in H^2(T^2;A) = A\big/\big((\sigma{-}1)A{+}(\tau{-}1)A\big), \qquad [x,y] = xyx^{-1}y^{-1} \in \ker(E \to Q) = A,$$

independent of lifts, where σ, τ are the b-twisted actions of s, t on A. We instantiate it on both extensions.

Strictification end (Σ)**.** Band $\rho : \mathbb{Z}^2 \to \mathrm{Out}K$, lifts $A, B \in \mathrm{Aut}K$:

$$\delta(\rho) = \big[[A,B]\big] \in H^2(T^2;\mathrm{Inn}K).$$

Central end (Z)**.** Here the coefficient is $A = Z(K)$ and the action is *trivial*: for $z \in Z(K)$ and any lift $k \in K$ of $\bar{k} \in \mathrm{Inn}K$, $\bar{k} \cdot z = kzk^{-1} = z$ (central). Hence $(\sigma - 1)Z(K) + (\tau - 1)Z(K) = 0$ and

$$H^2(T^2; Z(K)) = Z(K) \quad \text{(trivial coefficients, no quotient)}.$$

An *inner band* $\beta : \mathbb{Z}^2 \to \mathrm{Inn}K$ (automatically valid: $\mathrm{Inn}K$ abelian $\Rightarrow$ any pair of images commutes) with lifts $k, k' \in K$ gives

$$[\mathfrak{g}] = \gamma(\beta) = \big[[k, k']\big] = [k, k'] \in Z(K).$$

This is exactly the registry's Face 2 central-extension lifting class (`lift-cocycle-derivation`, proved), now read through the same lemma.

Remark 1 (What "independence" means here)**.** δ and $[\mathfrak{g}]$ live in *different* coefficient groups ($\mathrm{Inn}K$ vs $Z(K)$) and arise from *different* band targets ($\mathrm{Out}K$ vs $\mathrm{Inn}K$); for a single physical gerbe the central stage is only reached *after* strictification succeeds, so within one gerbe the pattern ($\delta \neq 0$, $[\mathfrak{g}]$ defined) cannot co-occur *by construction* and the question would be empty. The theorem is therefore a statement about the *group* K — does K admit a coupling band at each end, independently? — and *not* about any single gerbe. This is the four-corner cross-tabulation, and it is what we decide.

3 Two lemmas

Lemma 2 (central end is universally obstructed)**.** *For K finite of class ≤ 2: $[\mathfrak{g}] \neq 0$ for some inner band $\iff$ K is nonabelian.*

Proof. ($\Leftarrow$) K nonabelian $\Rightarrow \exists\, k, k' \in K$ with $[k, k'] \neq 1$. By class ≤ 2, $[k, k'] \in [K, K] \subseteq Z(K)$, so $[k, k'] \in Z(K) \setminus \{1\}$. Their images $\bar{k}, \bar{k}' \in \mathrm{Inn}K$ commute (abelian), so $\beta : s \mapsto \bar{k},\ t \mapsto \bar{k}'$ is a band, and by §2 $\gamma(\beta) = [k, k'] \neq 0$ in $H^2(T^2; Z(K)) = Z(K)$. ($\Rightarrow$) If K abelian then $[k, k'] = 1$ for all k, k', so $\gamma(\beta) = 0$ for every band. □

Lemma 3 (strictification end needs a nonabelian group)**.** *For any finite K: $\delta(\rho) \neq 0$ for some band ρ $\Rightarrow$ K is nonabelian.*

Proof. Contrapositive. If K is abelian then $\mathrm{Inn}K = 1$, so (Σ) reads $1 \to 1 \to \mathrm{Aut}K \xrightarrow{=} \mathrm{Out}K \to 1$: the kernel is trivial, every band lifts uniquely, and $\delta \equiv 0$ (the lemma gives $[A, B] \in \mathrm{Inn}K = 1$). □

4 Theorem: forced correlation

Theorem 4 (forced correlation)**.** *Let K be finite of nilpotency class ≤ 2. Then*

$$\big[\ \exists\, \rho : \delta(\rho) \neq 0\ \big] \implies \big[\ \exists\, \beta : [\mathfrak{g}] = \gamma(\beta) \neq 0\ \big].$$

Equivalently, the corner (δ couples, $[\mathfrak{g}] = 0$ for all bands) is empty*. The implication is* strict*: its converse fails — $K = D_4$ has $[\mathfrak{g}] \neq 0$ but $\delta \equiv 0$.*

Proof. If $\delta(\rho) \neq 0$ for some ρ, Lemma 3 gives K nonabelian, and then Lemma 2 gives a band β with $[\mathfrak{g}] = \gamma(\beta) \neq 0$. Strictness: for $K = D_4$ (dihedral of order 8, class 2), $\mathrm{Inn}D_4 = (\mathbb{Z}/2)^2$, and the extension (Σ), $1 \to (\mathbb{Z}/2)^2 \to \mathrm{Aut}D_4 \to \mathrm{Out}D_4 \to 1$ with $\mathrm{Aut}D_4 \cong D_4$, $\mathrm{Out}D_4 \cong \mathbb{Z}/2$, *splits*: the automorphism $\phi : r \mapsto r^{-1},\ s \mapsto rs$ is an order-2 element of $\mathrm{Aut}D_4$ that is *not* inner (it lies outside the inner Klein four-subgroup $\{1, r^2, s, r^2 s\}$), so $\langle \phi \rangle \to \mathrm{Out}D_4$ is a homomorphic section — hence $\delta \equiv 0$. Yet D_4 is nonabelian, so by Lemma 2 $[\mathfrak{g}] = [r, s] = r^2 \neq 0 \in Z(D_4)$. □

The structural cause (the asymmetry). The two "split?" questions underlying the four corners are *not* logically free:

- (Z) **is non-split for every nonabelian class-2** K**.** A splitting $\mathrm{Inn}K \to K$ would be a homomorphic section with image a complement S to $Z(K)$; because (Z) is central, $K = Z(K) \times S$ with $Z(K)$ and $S \cong \mathrm{Inn}K$ both abelian, forcing K abelian — a contradiction. (Coupling is then supplied directly by Lemma 2, using the trivial center-action.)
- (Σ) **can split** ($D_4, Q_8, \mathrm{Heis}(\mathbb{F}_3), M_{16}, C_4 \rtimes C_4$ all do), and split $(\Sigma) \Rightarrow \delta \equiv 0$ (a homomorphic section $\theta : \mathrm{Out}K \to \mathrm{Aut}K$ turns any band ρ into a homomorphic lift $\theta\rho$).

So among class-≤ 2 groups the column "(Z) splits" of the cross-tabulation contains *only abelian groups*; the central end is forced on for all nonabelian such K, while the strictification end is free. That collapse of one column *is* the forced correlation. (The "(Z) split $\Rightarrow$ abelian" step is a *nilpotency* phenomenon and fails outside it; see §8.)

Remark 5 (extension to all finite nilpotent K; attributed to R.)**.** The class-≤ 2 hypothesis above is a convenience, not the true domain. The following argument, which we owe to R. (referee), shows the Theorem holds *verbatim* for every finite nilpotent K. Let c be the nilpotency class and $\gamma_c = [\gamma_{c-1}, K] \neq 1$ the last nontrivial lower-central term. Since $\gamma_c \subseteq Z(K)$ and γ_c is generated by commutators, there exist $x, y \in K$ with $[x, y] \in \gamma_c \setminus \{1\} \subseteq Z(K) \setminus \{1\}$. Their images $\bar{x}, \bar{y} \in \mathrm{Inn}K$ commute (as $[x, y]$ is central, $\bar{x}\bar{y} = \overline{xy}$ and $\overline{xy} = \overline{yx}$ up to the central factor $[x, y]$, which acts trivially by conjugation), so $\beta : s \mapsto \bar{x},\ t \mapsto \bar{y}$ is an inner band and the central-end class is $\gamma(\beta) = [x, y] \neq 0$. Thus K nonabelian $\Rightarrow$ $[\mathfrak{g}]$ couples holds for all nilpotent K (not only class ≤ 2), (Z) remains a central extension with trivial coefficients, and the δ-end is Giraud's nonabelian obstruction on T^2 (no commuting lift pair), still computable. Composing with Lemma 3 (which assumed nothing about the class) gives the implication $\delta \Rightarrow [\mathfrak{g}]$ for all finite nilpotent K. We keep the class-≤ 2 proof above as the case originally established here; the nilpotent extension is R.'s.

5 Computational cross-tabulation (confirmatory)

The script `2026-10-13-two-ends-crosstab.py` computes, for each battery group: $|Z|, |\mathrm{Aut}|, |\mathrm{Inn}|, |\mathrm{Out}|$; whether $\mathrm{Inn}K$ is abelian (class ≤ 2); whether (Σ) splits and whether it δ-couples (exhaustive search for a commuting outer pair with no commuting lift); whether (Z) splits and whether it $[\mathfrak{g}]$-couples.

K	class ≤ 2?	(Σ) split	δ couples	(Z) split	$[\mathfrak{g}]$ couples
D_4	yes	**yes**	no	no	**yes**
Q_8	yes	yes	no	no	yes
$\mathrm{Heis}(\mathbb{F}_3)$	yes	yes	no	no	yes
M_{16}	yes	yes	no	no	yes
$C_4 \rtimes C_4$	yes	yes	no	no	yes
$D_4 \times C_2$	yes	no	**yes (30)**	no	yes
$Q_8 \times C_2$	yes	no	**yes (54)**	no	yes
SD_{16}, D_{16} (class 3, nilpotent)	no	—	no	no	**yes**
C_4, $(\mathbb{Z}/2)^2$ (abelian)	yes	yes	no	**yes**	no

Four corners (δ couples) $\times$ ($[\mathfrak{g}]$ couples):

- (yes, yes): $D_4 \times C_2$, $Q_8 \times C_2$.

- (yes, no): **EMPTY** — the forced-correlation corner.
- (no, yes): $D_4, Q_8, \mathrm{Heis}(\mathbb{F}_3), M_{16}, C_4 \rtimes C_4$ (every split-(Σ) nonabelian class-2 group), together with the class-3 nilpotent SD_{16}, D_{16}.
- (no, no): abelian K only (vacuous).

Every nonabelian row has (Z) non-split and [$\mathfrak{g}$] coupling; every abelian row has (Z) split and no coupling — exactly Lemmas 2–3. The decisive (yes, no) corner is empty, confirming the Theorem. (SD_{16}, D_{16} have Inn nonabelian (class 3), so the class-≤ 2 proof does not apply directly; but they are *nilpotent*, and both ends are perfectly well-defined and computable there: [$\mathfrak{g}$] couples (yes) — for D_{16}, $[r, s] = r^2$ is not central but $[r^2, s] = r^4 \neq 1$ is, the general γ_c-mechanism of Remark 5 — while δ does not couple (no). They thus sit in the (no, yes) corner, consistent with the nilpotent extension and with no violation of the empty corner.)

6 Verification (hostile-referee pass)

- **§2 central-end identification.** Trivial center-action is immediate (z central). The lemma applies verbatim to any abelian-kernel extension; (Z) qualifies with $A = Z(K)$. $H^2(T^2; Z(K)) = Z(K)$ for trivial coefficients is standard ($T^2 = K(\mathbb{Z}^2, 1)$, $H^2(\mathbb{Z}^2; M) = M$). The class is the bare commutator — matches `lift-cocycle-derivation` and the D_4 arithmetic $[r, s] = r^2$.
- **Lemma 2.** The class-≤ 2 hypothesis enters at one point: to land $[k, k']$ in $Z(K)$ directly, so that the *abelian* hinge $\mathrm{Inn}K = K/Z(K)$ makes any pair of images commute and $[k, k']$ is read as a central-end class. This is a convenient sufficient condition, *not* a necessary one: see the attributed extension to all finite nilpotent K in Remark 5 above, where centrality of a nontrivial commutator is recovered from the lower-central series rather than assumed.
- **Lemma 3.** Elementary; $\mathrm{Inn}K = 1 \Rightarrow \delta \equiv 0$.
- **Theorem 4.** Pure composition of the two lemmas; the empty corner is their conjunction. The structural-cause paragraph (central extension of a nonabelian class-2 group never splits) is an independent elementary check, and matches (Z)-split=False on every nonabelian computed row.
- **Strictness witness D_4.** $\mathrm{Aut}D_4 \cong D_4$ splits over the inner Klein four via the non-inner order-2 automorphism $\phi : r \mapsto r^{-1},\ s \mapsto rs$; verified by the engine (Σ split=True, δ-couple=False) and by hand. D_4 is the dihedral $4 \mid n$ instance at $n = 4$.
- **(Σ)-split check for $\mathrm{Heis}(\mathbb{F}_3)$.** $|\mathrm{Out}| = 48$, too large to read off by hand; the split was verified through $\mathrm{AutHeis}(\mathbb{F}_3) \cong C_3^2 \rtimes \mathrm{GL}(2, 3)$ (order 432, with $\mathrm{Inn} \cong C_3^2$), in which a complement to Inn is exhibited by the $\mathrm{GL}(2, 3)$ factor, giving a homomorphic section of π.
- **Boundary/degenerate cases.** Abelian K: both ends trivial (corner (no, no)). Rank-1 base S^1: $H^2(S^1; -) = 0$, both ends vanish, no content. Trivial band: commutator = 0, both obstructions 0.
- **Scope honesty.** SD_{16}, D_{16} (both of order 16) have Inn nonabelian (class 3), so the *class*-≤ 2 proof does not cover them directly; but they are *nilpotent* — the 2-power dihedral group D_{2^k} has class $k-1$ — so the attributed extension (Remark 5) does cover them, and both ends are computed (δ no, [$\mathfrak{g}$] yes) with no violation. The wider dihedral family D_n with $4 \mid n$, $n > 4$,

is *not* uniformly nilpotent: D_{12} and D_{20} (orders $12, 20$) are non-nilpotent, hence outside Remark 5, yet they too compute as (δ no, $[\mathfrak{g}]$ yes) and so *happen* to satisfy the implication. The genuine failures lie strictly beyond nilpotency (§8).

7 Scope and (non-)gaps

- **Proved domain = class ≤ 2; the true boundary is nilpotency.** The proof above is written for class ≤ 2 because then $\mathrm{Inn}K$ is abelian, both obstructions are ordinary abelian H^2 ($\delta \in H^2(T^2; \mathrm{Inn}K)$, (Z) a genuine central extension with $[k, k'] \in Z(K)$), and Lemma 2 is immediate. This is *not* the maximal domain: the implication in fact holds verbatim for *every* finite nilpotent K (see Remark 5, an argument we attribute to R.). The genuine dividing line is **nilpotency**, not class ≤ 2; it is identified, and shown sharp, in §8.
- **Grade.** The Theorem (forced correlation) is *proved* — it rests only on the proved δ-commutator lemma and two elementary lemmas. The battery cross-tabulation is *computed* over the enumerated family (not a census of all groups of each order; minimality is not claimed).
- **Not inflated.** The claim is a one-directional implication with a strict counterexample to its converse, *not* a biconditional and *not* a census theorem. The decisive content is the forced-empty corner plus its structural cause (the (Z)-split column is abelian-only).

8 The true boundary: nilpotency

The implication is *not* unconditional: it fails as soon as K is non-nilpotent. The cleanest witness is $K = \mathrm{Dih}(C_3^2) = C_3^2 \rtimes C_2$ of order 18, which is *centreless* ($Z(K) = 1$). Here $\mathrm{Aut}K \cong C_3^2 \rtimes \mathrm{GL}(2,3)$ (order 432), $\mathrm{Inn}K \cong K$, and $\mathrm{Out}K \cong \mathrm{PGL}(2,3) \cong S_4$. The normal $V_4 \subset \mathrm{PGL}(2,3)$ lifts to $Q_8 \subset \mathrm{SL}(2,3)$, so lifts of a commuting V_4-pair *anticommute*: the δ-end couples. (The cross-tabulation script finds 12 unordered Out-pairs that δ-couple.) But $Z(K) = 1$ forces $[\mathfrak{g}] \equiv 0$. Hence the corner ($\delta \neq 0$, $[\mathfrak{g}] = 0$) — which the Theorem forces empty in the nilpotent world — is *occupied* for this non-nilpotent K. Controls: S_3 and A_4 (also centreless, with cyclic Out) do *not* δ-couple, so centrelessness alone does not produce a violation — the δ-end still needs the right Aut-structure. (The $\mathrm{Dih}(C_3^2)$ witness, the count of 12 unordered δ-coupling Out-pairs, and the S_3, A_4 controls are due to R. (Rick), 2026-10-14.)

Correcting the structural cause. This pins down what the "(Z)-split column is abelian-only" statement of §4 really depended on. "(Z) splits $\Rightarrow$ K abelian" is a *nilpotency* fact, *not* a general one: for a centreless K the sequence (Z) reads $1 \to 1 \to K \xrightarrow{\sim} \mathrm{Inn}K \to 1$ and splits *trivially*, yet K is nonabelian. So the central end can be switched *off* while the strictification end couples precisely when *no nontrivial commutator is central* — which is exactly what nilpotency prevents, via the γ_c-central mechanism of Remark 5. (Centre size is not the parameter: R.'s $K = \mathrm{Dih}(C_3^2) \times C_2$ has $Z(K) = C_2 \neq 1$ yet still occupies the forbidden corner, because its commutators lie in the centreless $\mathrm{Dih}(C_3^2)$ factor.) Nilpotency is therefore the honest boundary of the forced correlation.

Remark 6 (open lead: no class-≥ 3 δ-coupling witness yet)**.** The nilpotent extension (Remark 5) is true but, at class ≥ 3, so far *vacuous as to δ-coupling*: in the small-order battery ($D_{16}, SD_{16}, Q_{16}, D_{32}, SD_{32}, Q_{32}, D$
$C_2, Q_{16} \times C_2, SD_{16} \times C_2, C_8 \rtimes C_4, C_3 \wr C_3, \dots$) *no* class-$\geq 3$ nilpotent group δ-couples. Thus every known δ-coupling nilpotent witness is class 2 ($D_4 \times C_2$, $Q_8 \times C_2$). Finding a class-≥ 3 nilpotent K

that δ-couples (or proving none exists) is open; the natural place to look is the $D_4 \times C_2$ mechanism — central automorphisms arising from $\mathrm{Hom}(K, Z(K))$.

One-line summary. On T^2, the strictification obstruction $\delta \in H^2(T^2; \mathrm{Inn}K)$ and the central obstruction $[\mathfrak{g}] \in H^2(T^2; Z(K))$ of a class-≤ 2 inventory gerbe are *not independent*: δ-coupling forces K nonabelian, and every nonabelian class-2 group has $[\mathfrak{g}] \neq 0$ (central extension non-split, trivial center-action), so $\delta \neq 0 \Rightarrow [\mathfrak{g}] \neq 0$ — a strict one-way forced correlation **(B)**; the converse fails at $K = D_4$ ($[\mathfrak{g}] \neq 0$, $\delta \equiv 0$, the dihedral $4 \mid n$ witness), and the corner $(\delta \neq 0, [\mathfrak{g}] = 0)$ is forced empty. The same holds for every finite *nilpotent* K (Remark 5, attributed to R.), and nilpotency is sharp: the centreless $K = \mathrm{Dih}(C_3^2)$ occupies the forbidden corner. The content is modest — the central end only detects nonabelianness — so the live question is which groups δ-couple.